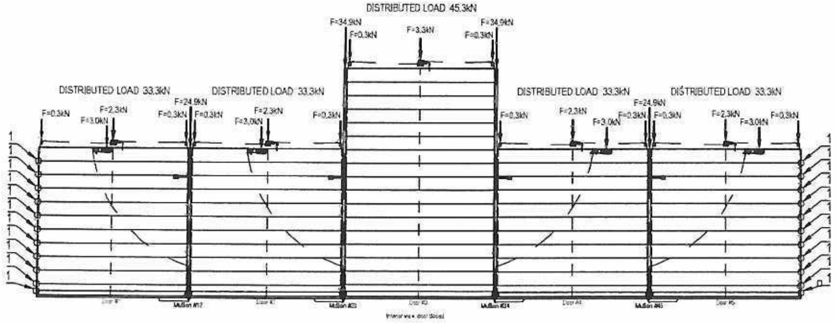
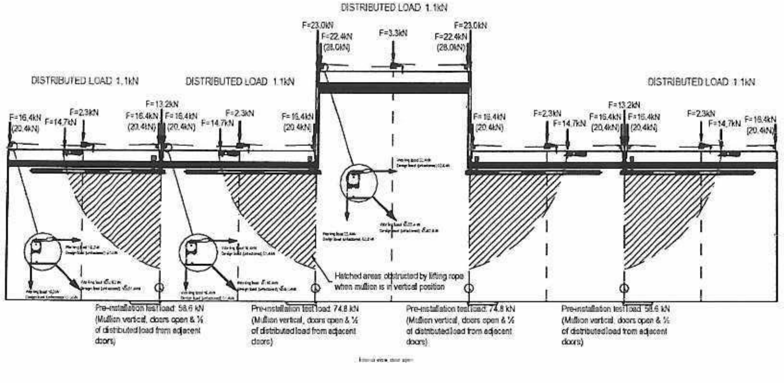
| Revision | Date | By |
|---|---|---|
| A Corrected mullion hoist positions, floorpits altered | 2022-08-02 | EB |

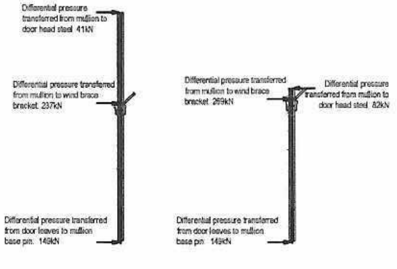
## LOAD DIAGRAM

Nominal static vertical loads are indicated. Value within brackets include dynamic surload factor 1,25 for door leaf loads in open position.
Includes items provided by ASSA ABLOY only.

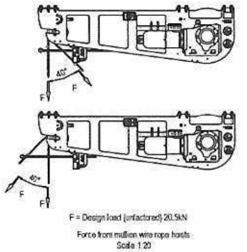




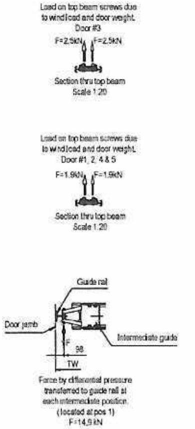
DISTRIBUTED LOAD 1.1kN
F=23.0kN
F=22.4kN
(28.0kN)
F=3.3kN
F=13.2kN
F=16.4kN
(20.4kN)
F=14.7kN
F=2.3kN
Hatched areas obstructed by lifting rope when mullion is in vertical position
Pre-installation test load 58.6 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)
Pre-installation test load 74.8 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)
Pre-installation test load 74.8 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)
Pre-installation test load 58.6 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)



Differential pressure transferred from mullion to door head steel 41kN
Differential pressure transferred from mullion to wind brace bracket 237kN
Differential pressure transferred from door leaves to mullion base pin 149kN
Differential pressure transferred from mullion to wind brace bracket 269kN
Differential pressure transferred from mullion to door head steel 82kN
Differential pressure transferred from door leaves to mullion base pin 149kN

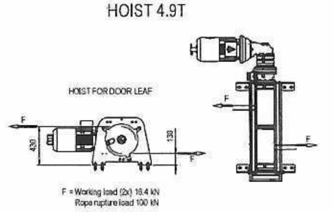


F = Design load (unfactored) 20.5kN
Force from mullion wire rope hoists
Scale 1:20

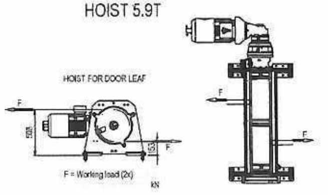


Load on top beam screws due to wind load and door weight. Door #3
F=2.5kN
F=2.5kN
Section thru top beam
Scale 1:20
Load on top beam screws due to wind load and door weight. Door #1, 2, 4 & 5
F=1.9kN
F=1.9kN
Section thru top beam
Scale 1:20
Guide rail
Door jamb
Intermediate guide
F
98
TW
Force by differential pressure transferred to guide rail at each intermediate position. (located at pos 1)
F=14.9kN

## HOIST 4.9T



## HOIST 5.9T



Steel building company in cooperation with door installing company must make a pre-installation load test on the structure, where the mullions are located, in order to simulate the vertical deflection of the structure that occurs when doors are installed. This vertical deflection figure will be used to determine the correct installation level of the mullions. Required material/equipment and personnel by others than ASSA ABLOY.

NOT INCLUDED IN DOOR SUPPLY
1. Design and supply of all door supporting structural components.
2. Design and supply of all closure between door and adjacent steel.
3. Design and supply of coverage, such as roofing or enclosure, for door components at door head.
4. Door jambs adjustable for overall door width and true to plumb.
5. Paint job after completed door installation.
6. Any frameworks being provided by ASSA ABLOY will be clearly indicated as "by ASSA ABLOY"

REQUIREMENTS ON BUILDING
7. Door head support must be straight within 1/800 of span in the in/out direction and free from protrusions of any kind.
8. For a single door leaf, door head support should be horizontal and straight within 1/600 of span at distributed full door load.
9. Door side jambs must be smooth, straight and parallel, out of plumb must not exceed 2mm/m (total deviation at full length 5 mm) in any direction.
10. Any footings at bottom of jamb columns must not interfere with installation of guide rails at floor level.
11. At time of installation finished floor free from obstructions, out to a distance of 2 m on each side of door centreline must be available.
12. Bracket/attachment plate for hoist should be flat within 5mm/m.

DOOR PROPERTIES
13. The door is designed to withstand a differential pressure of 1,2kPa.
14. Mullions and mullion pits are designed to allow for 133mm of downward deflection and 133mm of upward deflection due to external loads such as wind.
15. Guide rails top- and intermediate beams in extruded aluminium with natural finish.
16. Environmental category Megadoor standard
17. Bottom beam in carbon steel painted Light grey (RAL7047)
18. Colored damp strips on both sides
19. The door fabric is a UV stabilized, flame retardant, self-extinguishing, PVC coated polyester fabric with fire technical properties Euroclass B-s2, d0 (according to EN 13501-1:2007).
Colour of fabric TBC
RED (NCS S2070-R, RAL 3001)
BLUE (NCS S3560-R80B, RAL 5005)
GREY (NCS 3500, RAL 7004)
WHITE (NCS 0500, RAL 9016)
TAN (NCS 2010-Y40R, RAL 1001)
WHITE ALUMINIUM ( RAL 9006)
GREEN (NCS 8010-G10Y, RAL 6009)
ANTHRACITE GREY ( NCS 8005-B20G, RAL 7016)
(Please specify)

ELECTRICAL
20. One main control panel provided of PLC type.
21. One additional screen.
22. Control mode Impulse up / Constant contact down.
23. Cables are included.
24. Back-up power socket & switch.
25. Audible & visual alarm.
26. Approx. size of main control panel WxHxD (500x600x250 mm).
27. Location of main control panel. Please specify location.
28. Aux. Voltage 24VDC
29. Power supply 400V /3Ph /50Hz
30. Power 41.3kW
31. Nom. current 97.7A
32. Start current 214.2A
33. (max. allowed voltage drop at start 5%)

WIRE ROPE HOIST UNIT
LIDAN 4,9T2/27(4x)
2x Ø12mm wire ropes. Rupture load 100kN/rope.

LIDAN 5,9T2/27(1x)
2x Ø16mm wire ropes. Rupture load 178kN/rope.

MULLION HOIST UNIT
LIDAN 1.75T/1(4x)
2x Ø10mm wire ropes. Rupture load 69.8kN/rope.

DOOR IDENTITY NUMBER 84D0001

INTERFACE, SECTIONS ref. to drwng. no. 77445

INTERFACE, PLAN VIEWS ref. to drwng. no. 77444

INTERFACE, ELEVATION ref. to drwng. no. 77443

EB | A1F | 1:200
84600
2022-08-24

GMR Aero Technic Hangar1
India
Megadoor 84000x23000/14000
Load diagram & Notes

77446 | A

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